

Making do with less: Extending an acoustic-based time series of euphausiid abundance using an uncrewed surface vehicle with fewer frequencies

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ABSTRACT

Survey time series are used to track species- and ecosystem-level trends over time to support ecosystem-based fishery management. However, these recurring survey efforts are subject to unpredictable cancellations and reductions in effort. This occurred in 2020 when the COVID-19 pandemic forced the cancellation of the research vessel-based 2020 eastern Bering Sea (EBS) acoustic-trawl survey, which has provided an estimate of euphausiid abundance and distribution since 2004. As a partial replacement for this lost effort, three uncrewed surface vehicles (USVs) were used to collect acoustic data. In contrast to the standard vessel-based survey, which provides 4-frequency acoustic data, the USVs collected acoustic data at only two frequencies. This presented a challenge given that four frequencies are currently used to identify euphausiids in this time series. Here, we first evaluated two methods to provide comparable euphausiid abundance estimates using fewer acoustic frequencies. We found that a random forest classifier was able to produce abundance estimates comparable to those obtained in the vessel-based time series. This method was then used to estimate euphausiid abundance and distribution from the 2020 USV survey. We additionally estimated the increase in survey uncertainty due to the use of the random forest classifier and changes in the acoustic instruments. Together, this allowed for the EBS euphausiid abundance time series to be extended with fewer acoustic frequencies.

1. Introduction

Fisheries survey time series provide consistent information on species distribution and abundance, which is critical for stock assessment and management advice (Hilborn and Walters, 1992) as well as for tracking species- and ecosystem-level responses to sudden climatic events (e.g. the northeast Pacific marine heatwave; Harvey et al., 2020) and longer-term climate change (Pinsky et al., 2013; Currie et al., 2019; Gallo et al., 2022; O'Leary et al., 2022). While these time series are uniquely valuable, it is challenging to maintain them over the long term as surveys are vulnerable to cancellation or reductions in effort due to changes in funding, staff availability, equipment or vessel breakdowns, weather, or unexpected events such as the COVID-19 pandemic (Saunders et al., 2012; Harvey et al., 2020; ICES, 2020; Santora et al., 2021; Bryan and Thorson, 2023). Additionally, rapidly increasing industrial development such as offshore wind development (WFO, 2024) may exclude or limit survey activities in regions where fisheries survey extent overlaps with industrial activities (ICES, 2023; Lipsky et al., 2024). Given the many factors that can affect survey efforts, it will

remain challenging to provide consistent information over long time scales.

Autonomous platforms such as uncrewed surface vehicles (USVs), gliders, and moorings may provide a way to address some of the challenges faced in continuing long-term surveys. Autonomous platforms can allow for some types of data collection at a lower cost compared to research vessels which have traditionally been used to conduct many fisheries surveys (Verfuss et al., 2019; Handegard et al., 2024) and have relatively fewer operational constraints as compared to research vessels and at-sea staff. Given the reduced costs, several vehicles can be deployed to increase coverage and mitigate failures of a single vehicle (e.g. Greene et al., 2014; De Robertis et al., 2021).

Although there is substantial interest in integrating data from these platforms into survey time series to reduce costs and address logistical and staffing constraints, maintaining survey time series continuity with results obtained from traditional research vessel surveys may not be straightforward. For example, a multi-decade time series of *Euphausia superba* euphausiid abundance and distribution in the Southern Ocean, which was historically conducted as an acoustic-trawl survey from

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research vessels, was recently replaced with a glider-based survey effort without trawl sampling (Reiss et al., 2021). Comparisons with biomass estimates from traditional research vessels found broad agreement in one survey domain, but higher glider-based estimates in a second (Reiss et al., 2021), highlighting the challenges in maintaining consistency over time between different collection platforms.

In contrast to the Antarctic *E. superba* glider survey, which aimed to permanently replace a ship-based effort with an autonomous effort, other work has used autonomous platforms to provide information when a ship-based effort was unexpectedly suspended. In the eastern Bering Sea (EBS), the COVID-19 pandemic forced the one-time cancellation of a ship-based acoustic-trawl survey which has provided walleye pollock (*Gadus chalcogrammus*; hereafter referred to as pollock) abundance and distribution information since 1975 (Karp and Walters, 1994; Ianelli et al., 2022). In response, an acoustic-only survey with three Saildrone USVs (Saildrone, Inc.) was conducted in 2020 (De Robertis et al., 2021). Because pollock dominate backscatter in the deeper part of the water column in this low-diversity pelagic region (De Robertis et al., 2010; Honkalehto et al., 2011), trawling was not necessary to identify backscatter from pollock. However, the lack of trawling made it infeasible to produce size- or age-specific pollock biomass estimates (De Robertis et al., 2021). Despite this limitation, the USV survey was successfully used to estimate pollock abundance in a situation when other surveys were not feasible (Ianelli et al., 2020).

While the primary objective of the EBS acoustic time series is to monitor pollock abundance and distribution, it also provides multifrequency acoustic data that have been used to generate an index of euphausiid (primarily *Thysanoessa raschii* and *T. inermis*; hereafter referred to as ‘krill’) abundance extending back to 2004 (Ressler et al., 2012; Ressler, 2022). Krill are a major component of the EBS ecosystem, providing an important trophic link for fishes (Aydin and Mueter, 2007), seabirds (Hunt et al., 2002; Nishizawa et al., 2017), and marine mammals (Baraff, 2006; Witteveen et al., 2015). Thus, krill abundance is an important indicator of ecosystem status and prey availability and the EBS krill index is considered as part of the Alaska groundfish management process (Siddon, 2022). The EBS krill time series has also allowed for ecological investigations of the top-down (pollock predation) and bottom-up (temperature) influences on EBS krill population dynamics (Ressler et al., 2014; Hunt et al., 2016).

Given the potential value of the krill time series for ecosystem-based fishery management (Pikitch et al., 2004), an estimate was needed for 2020 that used acoustic data from the USV survey developed to estimate pollock abundance during the COVID-19 pandemic (Ianelli et al., 2020; De Robertis et al., 2021). The primary challenge in this application was the loss of multifrequency acoustic data: in contrast to the ship-based surveys, which make 18, 38, 120, and 200 kHz acoustic measurements, the Saildrone USVs only produced 38 and 200 kHz measurements.

The reduced number of acoustic frequencies was a key constraint for this application because the primary way to acoustically identify krill is via their frequency-dependent scattering response. Krill are amenable to identification using multifrequency methods at the frequency range (18 kHz – 200 kHz) available from research vessels: across this range, krill backscatter spans the transition from Rayleigh to geometric scattering (Holliday and Pieper, 1995) and is highly frequency-dependent (i.e. backscatter is stronger at higher frequencies than at lower frequencies). This pattern has long been exploited to identify krill in the Southern Ocean (Madureira et al., 1993; Lawson et al., 2008; Reiss et al., 2008). It is also fundamental to krill identification in the EBS and Gulf of Alaska, which builds on this relationship by first measuring acoustic frequency responses at 18, 38, 120, and 200 kHz (“dB-differencing”) and then comparing these to expectations from a library of trawl-verified samples (De Robertis et al., 2010; Ressler et al., 2012; Simonsen et al., 2016).

While different acoustic krill identification methods rely on the same general frequency response pattern, the specific choice of classification

parameters (e.g. the number of acoustic frequencies to include, or the magnitude of the frequency response required for an observation to be classified as krill) greatly affects the resulting krill abundance estimates (Jarvis et al., 2010; Jech et al., 2018). It is therefore imprudent to simply assume that replicating the current approach used in the EBS krill time series with fewer frequencies will produce comparable krill estimates. In other words, for the 2020 USV-based survey estimate to be useful in the EBS krill abundance time series, it needs to be comparable with the rest of the vessel-based time series despite the reduction in acoustic frequencies. Additionally, the degree to which uncertainty in the abundance estimate increases as a result of the reduction in acoustic frequencies must be quantified. In the current study, we first tested two different 2-frequency approaches to krill identification and compared results to those from the current 4-frequency method used in the EBS krill time series. We then derived an abundance estimate for the 2020 USV survey and accounted for the increased uncertainty in the 2020 estimate resulting from the reduction in available acoustic information.

2. Methods

2.1. Survey test data sets and acoustic data processing

To allow for comparison between krill identifications made using the 4-frequency Z-score method currently used in the EBS krill time series (Ressler, 2022) and 2-frequency alternatives, acoustic measurements were gathered from 10 previous acoustic-trawl surveys conducted from 2006 to 2022 (McCarthy et al., 2020). These surveys extended across the EBS shelf, sampling water depths from ~50–1000 m with transects spaced every 20 nmi (37 km; see McCarthy et al., 2020, their Fig. 1); the only exception to this was in 2022 when transects were instead reduced to 40 nmi due to unexpected vessel staffing shortages. Acoustic data were recorded to at least 500 m depth at 18, 38, 120, and 200 kHz during each survey. Standard sphere acoustic calibrations (Demer et al., 2015) were performed before and after each survey.

For each survey, the mean volume backscattering coefficient (S_V , dB re 1 m^{-1}) at each frequency was averaged into 5-ping horizontal by 5-m vertical cells and exported using Echoview software (V13.0, Echoview Software Pty Ltd). The signal to noise ratio (SNR) was calculated for the 120 and 200 kHz measurements (De Robertis and Higginbottom, 2007), and, to avoid biasing comparisons with background noise, only measurements with $\text{SNR} \geq 10$ dB were retained. At 18 kHz, ringing associated with the transmit pulse resulted in consistent noise that decayed with range. We estimated this noise for each survey and only retained 18 kHz measurements with an observed signal level ≥ 10 dB above the ringing noise (Ressler et al., 2012). The 38 kHz measurements were assumed to exhibit negligible levels of background noise for this application at surveyed depths of < 500 m based on passive noise measurements and visual inspection of echograms (Ressler et al., 2012). Cells that had, at a minimum, valid S_V measurements at 38 kHz, 120 kHz, and 200 kHz (i.e. $\text{SNR} \geq 10$ dB at all frequencies) were retained for subsequent analysis. This was done so that comparisons could be made in areas where both the existing method used in the time series and the 2-frequency 38/200 kHz methods from the Saildrone were possible. Finally, only acoustic observations during daylight hours were retained due to krill diel vertical movement (Mauchline, 1980), which can cause krill to move above the minimum detection depth of the echosounders and/or disperse into low-density aggregations during darkness and bias estimates low (Ressler et al., 2012).

2.2. Reference 4-frequency krill identification method

The goal of the current study was to produce krill backscatter estimates from the USV survey using the available frequencies (38 and 200 kHz) that were comparable to those produced using the current method (hereafter referred to as the ‘reference method’). The reference method is fully described in De Robertis et al. (2010) and Ressler et al.

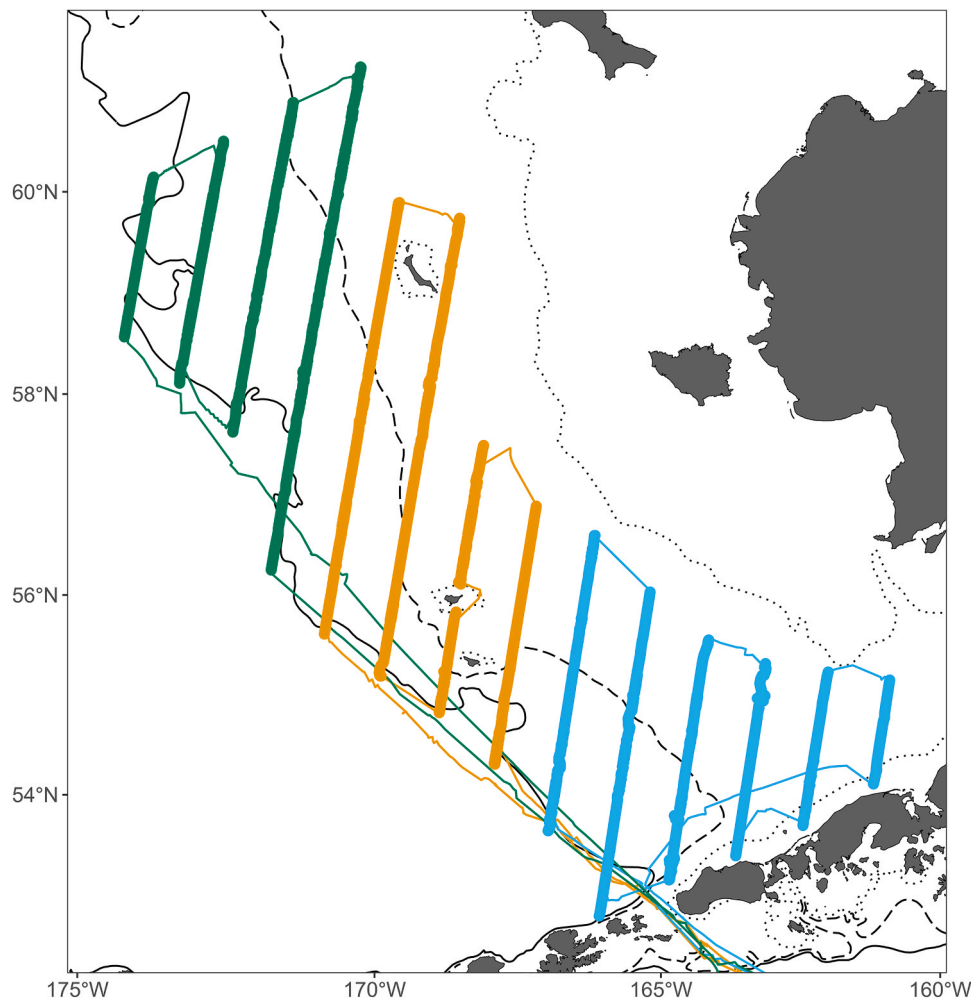


Fig. 1. Survey overview map. Each USV (represented by different line colors) surveyed parallel transects spaced 74 km apart. Wide lines indicate the path of the USVs along survey transects, while narrow lines indicate the path of the USVs to and from the survey transects. The 200 (—), 100 (—), and 50 (••) m depth contours are shown in black.

(2012), and a brief summary is presented here. Pairwise frequency differences are calculated for S_V measurements made at 18, 38, 120, and 200 kHz (i.e. ΔS_V , where for any given frequency pair i, j , $\Delta S_{V,i,j} = S_{V,i} - S_{V,j}$; De Robertis et al., 2010) within 5-ping horizontal by 5-m vertical cells. This yields 6 ΔS_V measurements in a cell where all four frequency measurements are available. The reference method allows for cases in which either the 18 kHz and/or the 200 kHz measurement is missing to be included, so in practice it is possible for fewer than six ΔS_V measurements to be present within a given cell (18 kHz measurements may be missing because they exceed SNR criteria due to ringing from the transmit pulse, largely at short range, and 200 kHz measurements may be missing due to a range-dependent increase in background noise, especially below 200 m). ΔS_V measurements are then compared to expectations based on trawl-verified samples for a number of taxa (pollock, krill, jellyfish, and myctophids), resulting in measurements of the deviation from the expectation for each ΔS_V value and taxon (referred to as 'Z-scores'). The mean Z-score for each cell relative to each taxon is then calculated, and cells for which the comparison to krill produces both the smallest Z-score and is ≤ 2 relative to krill are identified as krill (this is interpreted as a cell with a backscatter frequency response that is within two standard deviations of the expected frequency response for krill). Backscatter in cells identified as krill is integrated at 120 kHz with a -80 dB re 1 m^{-1} integration threshold into larger 0.5 nmi (926 m) horizontal $\times$ 20 m (vertical) elementary distance sampling units (EDSUs, Simmonds and MacLennan, 2005), resulting in a

measurement of the nautical area scattering coefficient (s_A , $\text{m}^2 \text{ nmi}^{-2}$; MacLennan et al., 2002) for each EDSU.

The average Z-score of all krill cells within each EDSU ($\bar{Z}_{krill}$) is calculated to provide a measure of how well the average frequency response of the cells identified as krill fits with the expected frequency response from krill (De Robertis et al., 2010). Only values from EDSUs with $\bar{Z}_{krill} \leq 1.5$ are retained as krill; this has the effect of removing outlier observations where the spatially-averaged response is unlikely to be from krill. Finally, a correction factor is estimated to address the potential for relatively weak krill backscatter to be 'shadowed' by relatively strong pollock backscatter when both taxa co-occur within an analysis cell. This effect is estimated by calculating the proportion of cells p where pollock are present in a local region (defined as a $5 \text{ nmi} \times 20 \text{ m}$ bin), and applying a correction factor to the observed krill backscatter in this region based on this proportion ($s_{A,corr} = s_A \times (1 + p)$; Ressler et al., 2012).

2.3. Testing and validation of alternative two-frequency krill identification methods

Two alternative 2-frequency identification methods were tested. The first method was the reference method, but using only the 38 kHz and 200 kHz measurements (referred to as the '2-frequency Z-score' method). This method is easily implemented, is analogous to the 2-frequency approach used widely in other studies (e.g. Lawson et al.,

2008; Phillips et al., 2022) and requires the fewest deviations from current practice. The only significant deviation was that we integrated backscatter in cells identified as krill at 200 kHz instead of at 120 kHz. This was done by integrating 200 kHz cells with a minimum integration threshold of $-77.7 \text{ dB re } 1 \text{ m}^{-1}$ (this threshold is equivalent to a -80 dB threshold at 120 kHz adjusted by the empirically derived mean $\Delta S_{V, 200-120}$ of 2.3 expected for krill; De Robertis et al., 2010). This modification allowed for comparisons of the 2-frequency Z-score method (which must be integrated at 200 kHz), and resulted in a measure of 200 kHz backscatter (s_A) within each EDSU.

The second method tested was to construct a random forest classifier (referred to as the 'RF method'). Random forest models are a collection of many classification trees, each created from a bootstrap sample of the available data and constructed using a random selection of variables from among the total available; the final classification is a simple 'majority vote' of the trees (Breiman, 2001; Hastie et al., 2009). These models are appealing because they are simple to implement and train, generally show low bias, and are not susceptible to overfitting (Hastie et al., 2009). In addition, assessing the performance of these models is relatively straightforward. The accuracy of a model can be assessed via cross-validation and/or by testing the model on withheld ('test') data (Hastie et al., 2009). The relative importance of each predictor to the classification can be assessed using variable importance metrics (Breiman, 2001).

The training data for the random forest model were taken from 10 previous EBS acoustic-trawl surveys. The krill identifications from the reference method were treated as 'correct' identifications, and the goal of the random forest classifier was to match these identifications. A series of simple predictor variables available from the 2020 USV survey data set were considered in model training: $S_{V,38 \text{ kHz}}$ and $S_{V,200 \text{ kHz}}$ for each 5-ping by 5 m cell, $\Delta S_{V,200-38 \text{ kHz}}$, cell depth, seafloor depth, distance above the seafloor, proportion of the water column depth at the cell location, cell latitude and longitude, and time of day. Given the reduced amount of acoustic information in comparison to the reference method, we hypothesized that adding positional and environmental variables in addition to the acoustic variables would increase the predictive power of the random forest classifier. Prior to inclusion in models, predictors were examined for collinearity using variance inflation factors (Fox and Monette, 1992). Although random forests are not overly sensitive to collinear variables, collinearity can make interpretation of the relative importance of predictors challenging (Breiman, 2001). Variables with variance inflation factors > 10 were therefore removed from models (Chatterje and Hadi, 2006; Naimi et al., 2014). This resulted in the removal of cell depth, the proportion of the total water column depth at the cell location, and $S_{V,38 \text{ kHz}}$ (given $S_{V,200 \text{ kHz}}$ and $\Delta S_{V,200-38 \text{ kHz}}$, $S_{V,38 \text{ kHz}}$ is known).

The total number of analysis cells for the 10 historical surveys was very large ($> 3.4 \cdot 10^7$). Including all of these samples in the training set was computationally impractical. To generate a reasonably-sized training set, all cells from $> 200 \text{ m}$ depth (the maximum depth at which USVs collected 200 kHz S_V measurements in 2020) were first excluded (in the 10 historical surveys, this excluded $5.7\% \pm 3.5\%$ of cells [mean $\pm$ SD]) from the training set). 10,000 cells were then selected at random from each survey, for a total of 100,000 cells in the training set. In addition, an additional 20,000 cells that were not included in the training set were randomly selected from each survey (200,000 cells total) to act as an independent estimate of classification error for the final random forest model. Classifications in the training dataset were not balanced: 25.4 % of the training cells were classified as krill, while the remainder were classified as non-krill (this reflects a typical percentage of krill cells in the 10 historical surveys, where the percentage of krill cells ranges from approximately 16–31 % of the total cells). While imbalances in the number of observations between classes can impact model performance, the observed ratio (1:4) is relatively balanced in comparison to levels that are generally considered to be substantially imbalanced (commonly $> 1:100$; Haibo and Garcia, 2009).

Additionally, the relatively large number of cells identified as krill in the training set ($> 25,000$ cells) ensures that krill identifications are not a rare case, and that there are many opportunities for krill identifications to be included in model training. Therefore, no efforts were made to balance the training dataset.

The random forest classifier was implemented using the R (version 4.2.3, R Core Team, 2023) packages *caret* (Kuhn, 2008) and *ranger* (Wright and Ziegler, 2017). The number of trees to construct (*ntrees*) was set at 500 as model accuracy tends to plateau before this point (Hastie et al., 2009; Probst and Boulesteix, 2018), and multiple values of the number of variables to randomly select from at each node (*mtry*) were tested (*mtry* = 2, *mtry* = 3, and *mtry* = 4); *mtry* = 2 was used for the final model because it performed marginally better than other values, but model performance was similar at all tested *mtry* values. Model performance was assessed using K-fold cross validation, in which folds were explicitly defined by year, with a single fold containing all of the observations from a given survey year ($n = 10$ folds total). In this validation approach, a model was created for year y using observations from all other years except y , and performance was tested on year y . This approach was used to evaluate how well the model prediction generalized to survey years that were not included in the training set.

Because the training data set was slightly imbalanced, we judge model performance using precision, recall, and the F1 score as opposed to accuracy (He and Ma, 2013). Precision measures how accurate positive predictions were, recall is a measure of how many of the total positives were correctly predicted, and F1 scores equally weight precision and recall to summarize overall model performance. These metrics are more appropriate than using accuracy, which simply measures the total number of correct positive (krill) and negative (not krill) predictions, because they specifically measure the performance of the less-common positive classification. We calculated precision as:

$$\text{precision} = \frac{\text{correctly predicted positive cells}}{(\text{correctly predicted positive cells} + \text{incorrectly predicted positive cells})}$$

recall as:

$$\text{recall} = \frac{\text{correctly predicted positive cells}}{\text{total number of positive cells in training set}}$$

and F1 as:

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

Overall random forest performance was assessed by the average precision, recall, and F1 scores in the 10 cross-validation folds as well as the kappa value (the proportion of classification agreement beyond what would be expected by chance; Cohen, 1960). The relative importance of each predictor variable was assessed by permutation (Breiman, 2001, 2002), where a given predictor was randomly reshuffled (permuted) and importance was defined as the increase in prediction error after permutation (an increase in prediction error would be expected after permutation of an important predictor). The final model used for krill prediction during the USV survey was fit using the complete training dataset (i.e. 10 survey years and 100,000 total observations).

To remove marginal krill classifications, both the reference method and the 2-frequency Z-score method require that cells identified as krill at the cell level be consistent with this expectation at a larger scale by requiring $\bar{Z}_{krill} \leq 1.5$ at the EDSU level. To emulate this approach in the random forest method, we require that krill identifications be spatially co-located with other krill identifications. This is in keeping with the observation that krill are distributed as spatially cohesive layers in the EBS (Ressler et al., 2012), and are therefore not likely to be observed as 'lone' cells. Cells identified as krill were required to be contiguous with ≥ 2 other cells identified as krill, using the chess

inspired ‘Queen’s case’ contiguity rule, where either cell corner vertices or sides that touch form a valid neighbor relationship. Krill that passed this criterion were then integrated within $0.5 \text{ nmi} \times 20 \text{ m}$ EDSUs at 200 kHz with an integration threshold of $-77.7 \text{ dB re } 1 \text{ m}^{-1}$, resulting in a measure of 200 kHz backscatter (s_A) within each EDSU.

To assess the accuracy of 2-frequency identifications at the survey level, total krill backscatter for the 2-frequency Z-score and RF methods was compared to values from the reference method. We express total krill backscatter within each survey and identification method as the total krill acoustic backscattering cross section (m^2) at 200 kHz. This is done by expanding s_A to the survey area:

$$k = \sum_i \frac{n s_{A,i}}{4\pi} E_i,$$

where E_i represents the survey area in nmi^2 represented by each 0.5 nmi EDSU i (20 nmi^2), and n the number of EDSUs.

2.4. USV survey

The Saildrone USVs, echosounders, and 2020 survey design are fully described elsewhere (De Robertis et al., 2021). Thus, only a brief summary is provided here. The survey was conducted during daylight hours only from July 4, 2020 – August 20, 2020 (for an overview of survey locations, see Fig. 1). The survey was designed to follow the design of the biennial EBS pollock survey (McCarthy et al., 2020) as closely as possible, but there are important differences between the 2020 USV effort and the ship-based surveys that comprise the remainder of the time series. The three Saildrone USVs were equipped with Simrad WBT (Kongsberg Simrad) mini echosounders and ES-38-18/200-18CR transducers (18° beam width split-beam 38 kHz and 18° beam width single-beam 200 kHz combined transducer), in contrast to the 11° beam width 18 kHz and 7° beam width 38, 120, and 200 kHz transducers used on the survey vessel. The USV survey covered the same spatial extent as the ship-based surveys, but transects were spaced every 40 nmi. This is half the 20 nmi resolution sampling of all EBS acoustic-trawl surveys, except for 2022 which also used 40 nmi transect spacing. Finally, 200 kHz Saildrone backscatter was recorded to only 200 m, in contrast to 500 m in the ship-based time series, which limited krill identifications to the upper 200 m of the water column. The effects of each of these differences on the 2020 USV survey krill estimate are addressed below.

2.5. USV acoustic data processing

Previous work with Saildrone USVs indicated that bubbles can be swept under the transducer during high wind and wave states which attenuates the acoustic signal, biasing backscatter observations low (De Robertis et al., 2019; Jech et al., 2021). This effect was mitigated by filtering out suspect pings with a weak bottom echo or disturbed transmit pulse prior to analysis (De Robertis et al., 2019, 2021), and pausing operations at wind speeds $> 12 \text{ m s}^{-1}$ (in practice, pauses were rarely required, as 95 % of wind speeds were $< 9.7 \text{ m/s}$ during the survey; De Robertis et al., 2021). $S_{V,38 \text{ kHz}}$ and $S_{V,200 \text{ kHz}}$ measurements were then averaged within 5-ping horizontal by 5-m vertical bins and exported using Echoview software. SNR was calculated for the 200 kHz measurements (De Robertis and Higginbottom, 2007), and 200 kHz measurements with $\text{SNR} \leq 10 \text{ dB}$ were excluded. As in the vessel-based surveys, 38 kHz measurements provided high SNR estimates of euphausiid backscatter at the surveyed depths, and no further SNR criteria were applied at this frequency. Measurements were then associated with a 0.5 nmi EDSU along the planned transect path. This was required because Saildrones rely on the wind to travel, so their course depends on wind speed and direction (e.g. a Saildrone cannot sail directly upwind along a transect and will cover a longer distance as it tacks upwind than one travelling downwind). Averaging all measurements within 0.5 nmi EDSUs along the intended transect removes this potential sampling bias (De Robertis et al., 2021).

Krill cells were then identified using the RF method and integrated into 0.5 nmi (horizontal) by 20 m (vertical) EDSUs along the intended transect at 200 kHz with an integration threshold of $-77.7 \text{ dB re } 1 \text{ m}^{-1}$. As in the reference method, a correction factor for pollock overlap (Ressler et al., 2012) was then applied to the EDSU-level krill values; this increased krill backscatter by 11.3 % survey-wide.

As previously noted, the Saildrone transducer had wider beam widths (18°) than the survey vessels (7°) at 38 and 200 kHz. The use of a wider beam transducer results in a larger acoustic deadzone near the sea floor where acoustic targets cannot be detected (Ona and Mitson, 1996). We estimated the unobserved backscatter attributable to the use of this wider-beam transducer by extrapolating the krill backscatter observed in the deepest cell of each 5-ping interval into the vertical zone observed by a 7° but not an 18° transducer and adding it to the observed krill backscatter in each EDSU (see De Robertis et al., 2021, their equation 1 for details). This correction depends on beam width, water depth, and the backscatter in the near-bottom layer (Ona and Mitson, 1996). Applying this correction increased total krill backscatter by 5.2 % survey-wide.

We also applied a correction for krill backscatter likely to be present below the 200 m Saildrone identification range using 10 surveys from the ship-based time series (which identified krill to 500 m). An average of 95.9 % of the total krill backscatter was detected in the upper 200 m (range 88.9 % - 99.5 % on individual surveys). The difference of this average value from 100 % was assumed to be the percentage of backscatter that would have been detected below 200 m in 2020 had the Saildrones been able to identify krill to 500 m, and the total backscatter estimate was inflated by 4.3 % (i.e. $(1/0.959)-1$) to account for the ‘missing’ observations below 200 m.

2.6. Converting backscatter to numeric density

The EBS krill time series is reported in units of krill m^{-3} . Conversion from backscatter to numeric abundance requires estimates of krill target strength (TS, $\text{dB re } 1 \text{ m}^2$; i.e. the acoustic backscatter from a single krill of a given size at a given acoustic frequency). We applied the numeric abundance conversion methods used in the EBS ship-based time series to the USV data. TS relationships from the existing EBS time series were estimated using a TS model (Smith et al., 2013) and length distributions from trawl samples from 2004 to 2016 (this dataset and TS model parameterization is described in Ressler, 2012, 2022). This approach relies on pooled trawl samples taken over the time series as opposed to using within-survey samples for each year due to the relatively small number of trawls in each survey (Ressler, 2022). This yielded two representative TS measurements at 120 kHz: $-90.4 \text{ dB re } 1 \text{ m}^2$ for krill in areas with bottom depths $> 100 \text{ m}$, and $-90.3 \text{ dB re } 1 \text{ m}^2$ for krill in areas with bottom depths $< 100 \text{ m}$, based on observed differences in krill size and species composition within each depth stratum. The EBS krill time series utilizes TS relationships that have been parameterized and validated at 120 kHz, but, as previously noted, the USV collected only 38 kHz and 200 kHz acoustic measurements. Therefore, $TS_{120 \text{ kHz}}$ was converted to $TS_{200 \text{ kHz}}$ by adding 2.3 dB to each $TS_{120 \text{ kHz}}$ measurement (the empirically derived difference for krill backscatter at 120 kHz vs 200 kHz; De Robertis et al., 2010), yielding two $TS_{200 \text{ kHz}}$ values: $-88.1 \text{ dB re } 1 \text{ m}^2$ for $> 100 \text{ m}$ bottom depths and $-88.0 \text{ dB re } 1 \text{ m}^2$ for $< 100 \text{ m}$ bottom depths.

TS was then converted into σ_{bs} (m^2 , the backscattering cross section of a single krill; that is, TS in linear units) within each depth stratum:

$$\sigma_{bs,s} = 10^{\left(\frac{TS_{200 \text{ kHz},s}}{10}\right)}.$$

Krill $S_{V,200 \text{ kHz}}$ measures were then used to estimate abundance in each EDSU i and depth bin j . $S_{V,200 \text{ kHz}}$ was converted to its linear equivalent, $s_{V,200 \text{ kHz}}$ (m^{-1}):

$$S_{V,200kHzij} = 10^{\left(\frac{S_{V,200kHzij}}{10}\right)}.$$

The maximum bottom depth within each EDSU i was then used to select the proper $\sigma_{bs,s}$ value, and the numeric density of krill (individuals m^{-3}) at EDSU i and depth bin j was then calculated as:

$$N_{ij} = \frac{S_{V,200kHzij}}{\sigma_{bs,s}}.$$

Finally, the average survey-wide krill density $\bar{N}$ was calculated by averaging krill density vertically within each EDSU i , and then averaging all EDSUs across the survey area.

2.7. Uncertainty estimation

In the EBS krill time series, a geostatistical measure of sampling uncertainty is used to estimate the coefficient of variation for survey estimates of krill density ('1-D CV'; Ressler et al., 2012). This metric is based on a method that evaluates the observed spatial structure of the population (Petitgas, 1993), but does not consider additional sources of uncertainty. For the 2020 Saildrone survey, we calculate a 1-D CV, as in Ressler et al. (2012), and also accounted for additional uncertainty introduced by the methodological changes associated with the USV survey: the use of the RF method, the use of the $TS_{120\text{ kHz}}$ to $TS_{200\text{ kHz}}$ conversion factor, and the correction factor for the unobserved deep ($> 200\text{ m}$) krill backscatter.

We accounted for these sources of uncertainty using a Monte-Carlo simulation approach similar to that used in the pollock abundance estimate for the 2020 USV survey (De Robertis et al., 2021). The starting value for this simulation was the total observed 200 kHz krill backscatter b in the 2020 survey (m^2 ; note that b includes the corrections for the acoustic deadzone, krill/pollock overlap, and the backscatter present below 200 m). For each of 10,000 realizations s' , b was multiplied by factors randomly drawn from normal probability distributions representing the additional variability associated with the USV survey:

$$s'_{200kHz} = s_{200kHz} \times U_{id} \times U_f \times U_a.$$

s_{200kHz} (a potential realization of the USV 200 kHz backscatter estimate b , given the sampling uncertainty) was calculated as

$$s_{200kHz} = N(b, b \times 1DCV),$$

where N represents a draw from a normal distribution with mean = b and SD = $b \times 1\text{-D CV}$.

U_{id} represents the additional uncertainty introduced by use of the RF method for krill identification. This was estimated by using the 10 EBS krill surveys in the training set to construct a linear regression with an intercept of zero of the predicted total krill backscatter from the reference method given the RF method krill backscatter (described in the results section). The standard error of the prediction interval ($se_{\text{prediction error}}$, the expected uncertainty in an individual future prediction) was used to estimate the krill backscatter expected from the reference method by drawing from a normal distribution with a mean corresponding to the slope of the regression line and a SD corresponding to CV of the prediction error:

$$U_{id} = N\left(\text{slope}, \frac{se_{\text{prediction error}}}{b}\right).$$

The average $\Delta S_{V,200-120\text{ kHz}}$ response was used to convert krill target strength from 120 to 200 kHz, but the estimates of this average value are subject to error. U_f represents the uncertainty associated with the TS conversion from 200 kHz to 120 kHz. This source of uncertainty was represented by applying the empirically-identified uncertainty in the average krill frequency response at this frequency pair as per De Robertis et al., 2010:

$$U_f = 10^{\left(\frac{N(0,0.27)}{10}\right)},$$

where $N(0, 0.27)$ reflects a draw from a normal distribution with mean = 0 and SD corresponding to the empirically observed standard error of $\Delta S_{V,200-120\text{ kHz}}$ (0.27 dB; see De Robertis et al., 2010, their table 2, but note that this table presents SD as opposed to SE).

The uncertainty associated with the correction for unobserved deep krill backscatter U_a was represented by

$$U_a = N(1, 0.04),$$

where N represents a normal distribution with mean = 1 and 0.04 represents the observed standard deviation of this value among the survey years.

After repeating this process 10,000 times, the final survey CV was calculated as

$$CV = \frac{SD_s}{\bar{s}},$$

where SD_s and $\bar{s}$ refer to the standard deviation and mean of all replicates of s'_{200kHz} respectively. This CV was used to calculate approximate 95% confidence intervals for krill density $\bar{N}$ (individuals $\cdot m^{-3}$) as in Walline (2007):

$$95\%CI = 2 \times CV \times \bar{N}.$$

3. Results

3.1. 2-frequency Z-score method

In comparison to the 4-frequency reference method, the 2-frequency Z-score method was consistently biased high (Fig. 2). While the 2-frequency Z-score method captured the same general trends in krill backscatter over time as the reference method (Fig. 2A), it produced higher krill backscatter values than the reference method (mean 18.2 % higher than the reference method, range 1.2 % - 61.2 %; Fig. 2B). This is evident in a regression of the 2-frequency Z-score method and the reference method: while estimates were strongly correlated ($r^2 = 0.99$), the slope of the regression was less than 1 (slope = 0.93, 95 % CI = 0.87 - 0.98), indicating that the 2-frequency Z-score method is a biased predictor of the reference method.

3.2. Random forest method

Preliminary models indicated that adding geographic location and time of day as predictors added very little to random forest model performance ($< 1\%$ increase in precision, recall, and F1 score). Given that these terms were not influential, they were removed so that the model would be simpler and more general. The predictors used in the final model were $S_{V,200\text{ kHz}}$, $\Delta S_{V,200-38\text{ kHz}}$, seafloor depth, and the distance of the cell above the seafloor. Random forest classifications in an out-of-sample survey produced few false positive identifications in the final model (precision = 85.6 %), and the majority of true positives were correctly identified (recall = 93.0 %). The random forest model F1 score was 89.1 % and the kappa score was 0.82. The 'leave-one-year-out' cross-validation approach indicated that precision declined in years where krill were comparatively rare (such as 2016 and 2018), but that recall was relatively stable regardless of relative krill abundance (Table S1). In other words, a higher proportion of random forest krill identifications were false positives when krill were scarce. The withheld test dataset produced similar results to the 'leave-one-year-out' cross-validation testing process: precision (85.9 %), recall (93.8 %), and F1 scores (89.7 %) were similar to the values obtained during cross-validation.

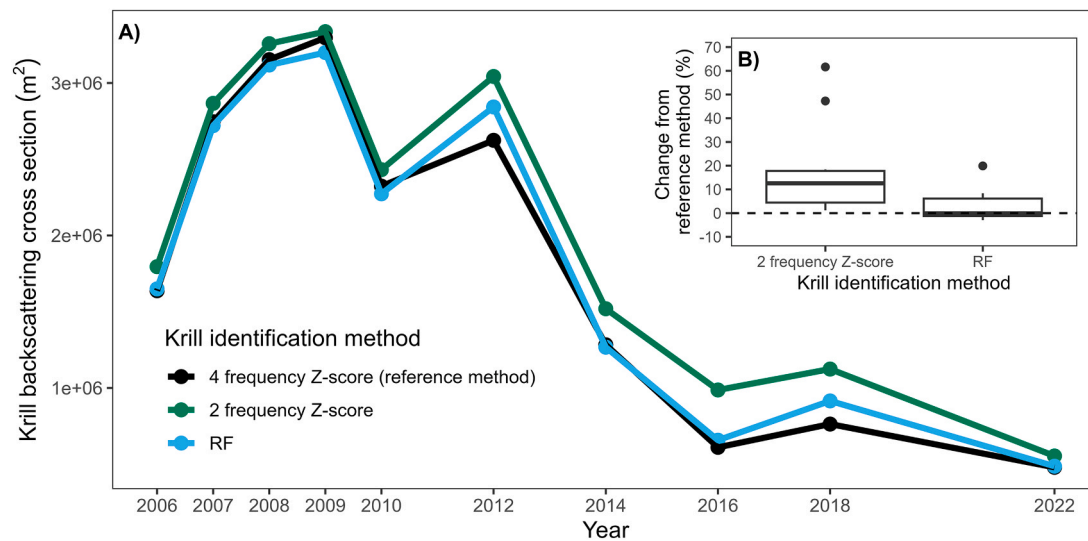


Fig. 2. Comparison of krill identification methods. A) total 200 kHz krill backscattering cross section per year from the 4-frequency Z-score reference method (black), the 2-frequency Z-score method (green), and the final RF model fit to all available data (blue). B) boxplots of the performance of two krill identification methods relative to the 4-frequency Z-score reference method. The solid line within each box represents the median difference from the reference method, the lower and upper limits of each box represent the first and third quartile, whiskers represent 1.5 times the interquartile range and dots represent outliers.

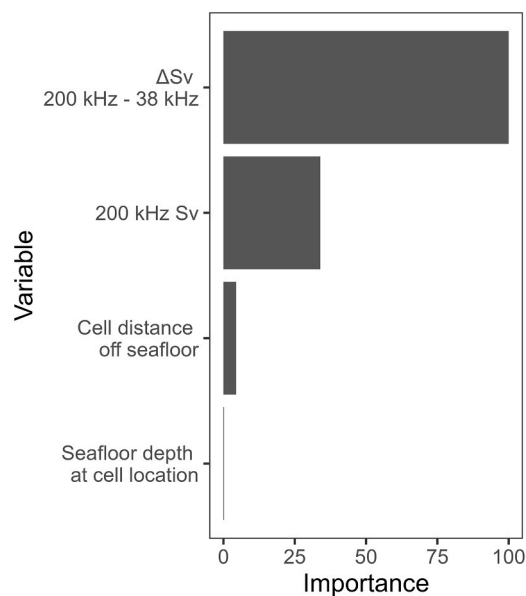


Fig. 3. Relative importance plot for the random forest classification model. Importance was determined by permutation and is scaled between 0 and 100.

Relative importance testing indicated that variables derived from backscatter ($\Delta S_{V,200-38 \text{ kHz}}$ and, to a lesser extent, $S_{V,200 \text{ kHz}}$) were much more important predictors than positional (cell distance off seafloor) or environmental (seafloor depth) variables (Fig. 3). Application of the ≥ 2 neighbors criterion had only moderate effects, indicating that most of the cells predicted to be krill by the RF method were spatially contiguous. In the 10-year time series, this criterion removed an average of $10.4 \pm 7.6 \%$ (mean $\pm$ SD) of cells initially identified as krill, and these cells comprised $10.1 \pm 9.6 \%$ of the backscatter initially attributed to krill.

In comparison to the 2-frequency Z-score method, the RF method produced less biased estimates of total krill backscatter that were much closer to those of the reference method (the RF method averaged 3.0 % higher than the reference method [range -2.9 – 19.9%]; Fig. 2). The estimates from the RF method were closely correlated to estimates to

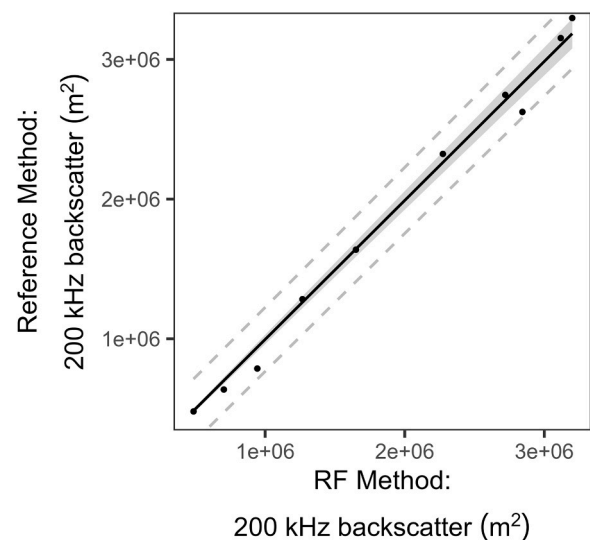


Fig. 4. Relationship between total krill backscatter as estimated by the 2-frequency RF method and the 4-frequency Z-score reference method ($n = 10$ surveys). The black line represents a linear regression with zero intercept, the grey ribbon represents the 95 % confidence interval for the regression, and the grey dashed lines represent the prediction intervals (i.e. 95 % confidence intervals for a future prediction).

those from the reference method ($r^2 = 0.99$; Fig. 4). This regression had a slope that was not significantly different from 1 (slope = 0.99, 95 % CI = 0.96 – 1.03). Together, this suggests that the RF method was an unbiased predictor of the reference method.

The RF method also identified similar krill spatial distributions as the reference method, both geographically and vertically within the water column (note that spatial distributions resulting from the 2-frequency Z score method were not assessed given that it proved to be a biased predictor of the reference method). Estimates of the center of gravity (a geostatistical measure of the mean location of a population; e.g. Woillez et al., 2007) for the RF method for each survey year were generally in close agreement to those from the reference method (mean distance from reference method $40.0 \text{ km} \pm 66.6 \text{ km}$; for context, the EBS survey

area covers an extent of approximately 1060 km length $\times$ 325 km width; Fig. S1). The average depth below the surface of krill within each survey year (weighted by backscatter) was 1.3 ± 3.4 m (mean $\pm$ SD) deeper in the RF method than the reference method (Fig. S2).

3.3. 2020 USV krill abundance estimate

Because the 2020 estimate was obtained using the RF method, as opposed to the reference method used in the remainder of the time series, it is informative to examine if the geographic and vertical distribution of the 2020 krill backscatter is consistent with the rest of the time series. Backscatter in the 2020 survey was largely concentrated in the southwest corner of the EBS shelf between 100 and 200 m, with low krill backscatter in the northern and offshore parts of the survey area (Fig. 5; see panel for 2020). This distribution is broadly consistent with those from the temporally adjacent survey estimates in 2018 and 2022 (Fig. 5, compare panels for 2018–2022). Based on center of gravity estimates, krill concentrations have shifted eastward from 2018 (where krill were concentrated in deeper water towards the shelfbreak) to 2022 (where krill were concentrated in shallower water), with the 2020 distribution centered between these two distributions (Fig. 5). The average

backscatter-weighted depth of krill observations in the 2020 saildrone survey (110.4 m) was 4.0 m deeper than the mean observation for the time series, and well within the range observed in the time series (94.0 m – 130.8 m, Fig. 6).

The 2020 krill density estimate was 31.6 ± 9.4 krill m^{-3} (mean, 95 % CI), a moderate value in the context of the krill time series (mean 24.7 krill m^{-3} , range $6.4 - 49.8$ krill m^{-3} ; Fig. 7). While the 2020 estimate suggests a continued trend towards increasing density since 2016 (the second lowest density in the time series), the 2020 estimate is not exceptionally high and is lower than those observed in 2007–2010.

The overall CV for the 2020 krill estimate is 14.9 %. This is nearly three times higher than the average value in the time series (mean = 5.1 %, range = 2.1 – 10.0 %). However, the uncertainty estimate for 2020 is based on fewer transects than most of the time series and also includes additional sources of uncertainty associated with the USV survey. Fig. 8 presents a visual representation of the Monte-Carlo resampling process, demonstrating the relatively large contribution of sampling uncertainty and RF method uncertainty to the total CV estimate. It also demonstrates the comparatively minor increase in uncertainty resulting from the TS conversion from 120 kHz to 200 kHz and the correction for deep unobserved krill. The 1-D CV of 6.1 % for 2020

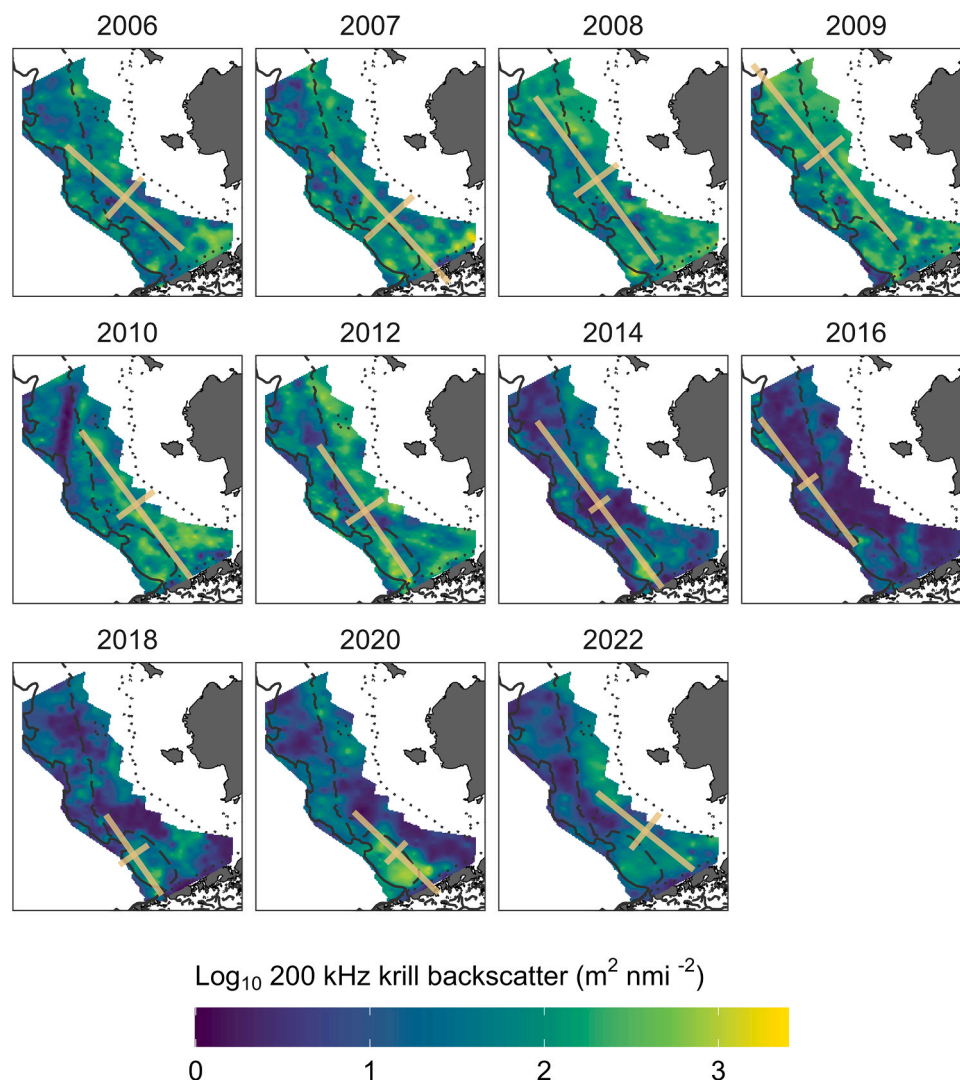


Fig. 5. Time series of 200 kHz backscatter measurements attributed to krill in the EBS. All years except 2020 represent shipboard surveys in which krill were identified using the 4-frequency reference method, while in 2020 3 USVs were used and krill were identified using only two frequencies with the RF method. Observations have been interpolated using universal kriging (Pebesma, 2004) within the survey area to make the distributions easier to visualize. Center of gravity and inertia (light crosses) are included to indicate the mean location and dispersion of krill backscatter within each survey year. The 200 (—), 100 (—), and 50 (••) m depth contours are shown in black.

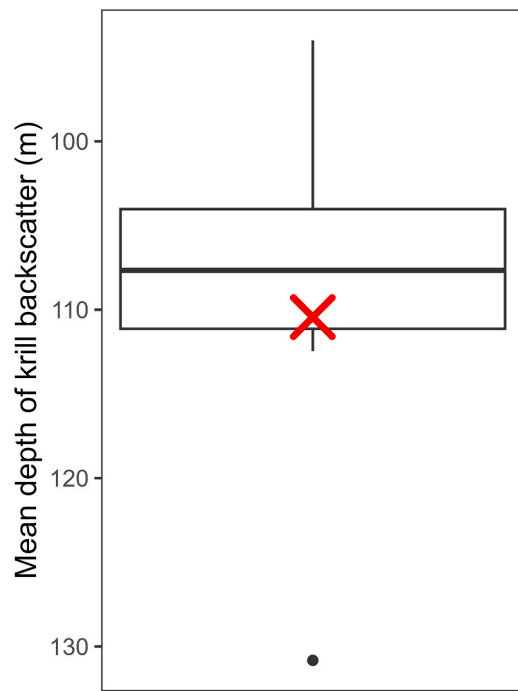


Fig. 6. Mean depth of krill backscatter (weighed by abundance) for 10 previous ship-based surveys (boxplot) and the 2020 USV-based survey (red X). The solid line within the boxplot represents the median depth of krill backscatter, the lower and upper limits of the box represent the first and third quartile, whiskers represent 1.5 times the interquartile range and dots represent outliers.

represents a relatively high value of sampling uncertainty in the time series, as would be expected given the 40 nmi transect spacing (2022, which had the highest 1-D CV in the ship-based time series, shared this transect spacing). Adding the uncertainty associated with the use of the RF method for krill identification increased the overall CV by 7.0–13.1 %. The uncertainty associated with the TS frequency conversion added an additional 1.3 % to the CV, while the correction for deep unobserved krill added an additional 0.5 %. Thus, the majority of the increased uncertainty in 2020 is attributable to sampling uncertainty and use of RF method.

4. Discussion

The random forest approach allowed for an unbiased EBS krill abundance estimate to be opportunistically developed from USV data after the sudden cancellation of a long-standing ship-based survey during a global pandemic in 2020. The primary challenge with the use of USVs rather than a research vessel in this application is both a reduction of acoustic information (ship-based observations are generally made to 500 m at 18, 38, 120, and 200 kHz, while the USVs only collected observations to 200 m at 38 and 200 kHz) and a loss of directed trawl sampling. In this study, we focused on addressing the reduction in acoustic information, and developed a method that produced results comparable to those of the 4-frequency method used in the EBS krill time series using only the two frequencies available from the USV. For the specific application of continuing the EBS krill time series, the loss of trawl sampling is not as problematic as the loss of acoustic information. While trawl sampling has been used to establish that krill have a consistent frequency response and is used in the time series to convert acoustic backscatter to krill abundance, the conversion has generally been based on an average catch over many survey years, and krill length estimates have been relatively consistent interannually (Ressler et al., 2012; Ressler, 2022). Given how the length estimates are used in this time series, the lack of trawl sampling data in a single year is not likely to impact our estimate relative to a ship-based estimate applying the same methodology. In contrast, the lack of 18 and 120 kHz backscatter measurements directly impacts krill identification.

The 2-frequency Z-score approach, which simply followed the reference approach (De Robertis et al., 2010; Ressler et al., 2012) with fewer frequency comparisons, was able to capture the trends in a 10-survey test dataset, but it produced estimates that were biased ~ 18 % high relative to the 4-frequency reference method. Previous work with multifrequency krill identification has also found that the use of fewer frequencies tends to produce relatively high krill abundance estimates. Jarvis et al. (2010) tested multiple dB-differencing approaches to krill identification in the Southern Ocean and found that using only two frequencies resulted in a biomass estimate that was nearly twice as high as the estimate obtained using three frequencies. Similarly, Simonsen et al. (2016) applied the reference method used in the current study with and without the 18 kHz measurement (i.e. as both a 4-frequency and a 3-frequency method) in the Gulf of Alaska, and found that removing 18 kHz measurements caused backscatter identified as krill to increase by ~ 30 % in 1 out of 4 years. In that single year, aggregations with a krill-like frequency response, except for relatively strong 18-kHz

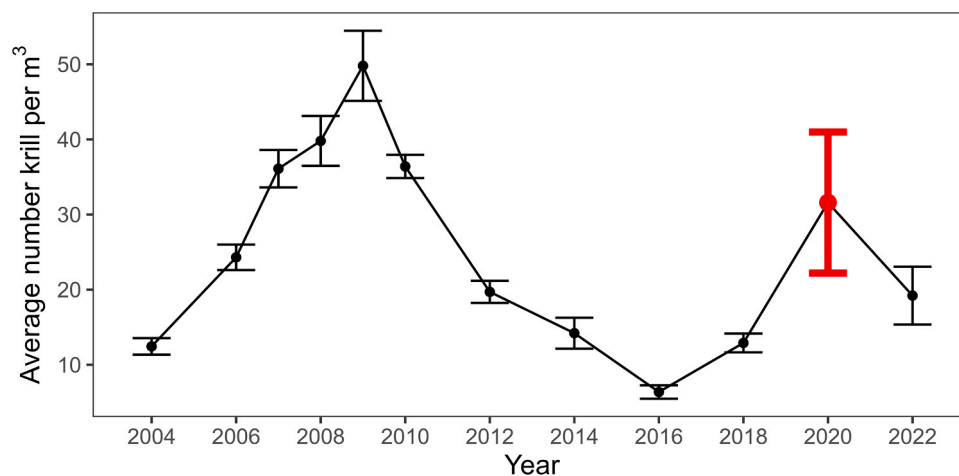


Fig. 7. Krill abundance index from 2004 to 2022. In 2020 (red), three USVs were used to conduct the EBS survey and krill were identified using the RF method using only two frequencies. All other years (black) represent ship-based surveys in which krill are identified using the 4-frequency reference method. Error bars in all years except 2020 represent approximate 95 % confidence intervals from a 1-D geostatistical estimate of sampling error, while the error bar in 2020 includes sampling error and additional uncertainties associated with use of USVs in this application (see text for details).

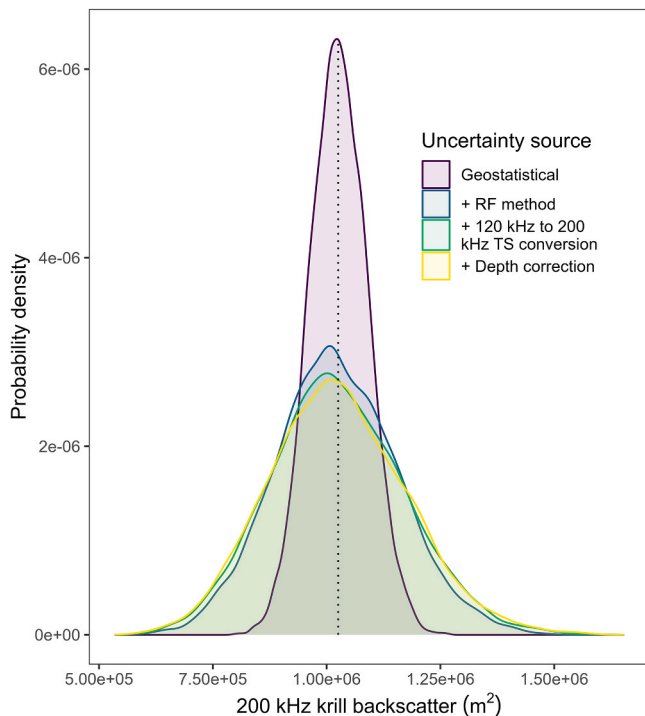


Fig. 8. Monte-Carlo error estimation simulation. The vertical dashed line represents the krill backscatter estimate from the 2020 Saildrone survey. Histograms represent the probability density of krill backscatter estimates ($n = 10,000$) after sequential addition of each uncertainty source (geostatistical, RF method, 120 kHz to 200 kHz TS conversion, and depth correction; see Methods for more detail) in the Monte-Carlo simulation.

backscatter, were present and identified as krill in the 3-frequency (but not the 4-frequency) method (Simonsen et al., 2016). Similarly, krill identifications in the Southern Ocean based only on two frequencies can mistakenly identify backscatter from fish without swimbladders as krill (Fallon et al., 2016). In an analogous situation where lesser sandeel (*Ammodytes marinus*) were identified using multifrequency methods, Komiya et al. (2024) found that acoustic identifications made with only 38 kHz and 200 kHz measurements were less accurate than those that also included an 18 kHz measurement. It is likely that as the number of frequency pairs for multifrequency identification is reduced, the identifications will become less robust, particularly if frequencies capturing important acoustic contrasts between scatterers become unavailable.

A random forest classification model trained to ‘mimic’ the 4-frequency identifications with two frequencies yielded estimates that were much closer to the 4-frequency method than the 2-frequency method. Overall, the random forest classifier appeared to use backscatter information in a way that was consistent with the known acoustic scattering properties of krill to make identifications comparable to those of the reference method. Relative importance testing of predictors indicated that the random forest classifier relied most heavily on the same information available to the 2-frequency Z-score approach, $\Delta S_{V,200-38 \text{ kHz}}$. This is in keeping with the known scattering characteristics of krill, which show a strong frequency response between 38 kHz and 200 kHz (Madureira et al., 1993). However, the absolute level of backscatter ($S_{V,200 \text{ kHz}}$) was also relatively important, which is consistent with the observation that krill are relatively weak scatterers. High backscatter is more consistent with aggregations of swimbladdered fish than krill in the EBS, and it is likely that the random forest classifier is better able to exclude fish aggregations with high scattering strengths based on this characteristic. In contrast, the addition of positional (latitude, longitude, and observation depth) and environmental (bottom depth) descriptors was relatively unimportant to the random forest classifier, which is

consistent with the relatively large amount of interannual variability in spatial distributions (Fig. 5). The importance of acoustic descriptors makes sense given the scattering properties of krill and their acoustic contrasts with other dominant scatterers in this environment (De Robertis et al., 2010; Ressler et al., 2012). These properties are unlikely to change dramatically on annual time scales, which likely explains why ‘leave-one-year-out’ cross validation found that the model generalized well across years. It is important to note that our study intentionally avoided using more complex positional (for example, proximity to important bathymetric features such as canyons or the shelf break) or environmental (for example, measures of primary productivity or water temperature) predictors that may correlate with euphausiid abundance as these relationships are complex and often vary between euphausiid species and among survey regions (Evans et al., 2022). In contrast, the relatively straightforward physics-based acoustic properties that the random forest classifier relied most heavily upon are more likely to remain consistent as they are linked to the size and body composition of these organisms (Lavery et al., 2007). However, in more complex cases where acoustic properties alone do not allow for acceptable classifier performance, it may be appropriate to include more complex predictors.

Random forest approaches may be especially well suited for ‘filling gaps’ in survey time series. Because the classifier can be trained on previous output from the standard method used in the time series, a high-performing model will, by definition, produce results that have been optimized to be similar to those produced by the standard method. The random forest approach, in this use case, can be thought of as a tool to deal with missing data or methodological changes rather than an improvement over an existing classification method. The approach is straight-forward to implement and can be applied to mitigate unexpected situations, for example, the malfunction of a shipboard echosounder (Saunders et al., 2012), or if a less-well instrumented ship or USV must be used to continue a survey time series (e.g. due to vessel breakdown or cost considerations). The price of using this approach is an increase in classification uncertainty: even a high performing classifier is unlikely to exactly reproduce a standard method that is based on more (or better) information, and it is informative to estimate how large this impact is. Our resampling approach suggests that using the random forest approach more than doubled the 2020 krill CV (from 6.1 % to 13.1 %).

Previous studies using random forest classifiers for fish and zooplankton identification have utilized complex descriptors of school morphology and location (such as those calculated by the SHAPES algorithm, Coetzee, 2000). These school-based metrics can add useful information to classification models, and have been as informative as acoustic descriptors (such as backscattering strength) in some cases (Fallon et al., 2016; Proud et al., 2020). In the current study, we instead rely on a simple spatial contiguity rule that is based on previous observations that krill are aggregated in continuous layers in the EBS during daytime rather than forming schools or swarms (Ressler et al., 2012). By avoiding the inclusion of complex geometric descriptors in our model, it is likely to remain more generally applicable. Given that the high performance of our model was driven mostly by frequency response, which is the basis of the reference method, the more general approach is appropriate.

While the random forest approach allowed krill backscatter to be identified in a manner consistent with an existing time series, there are other challenges associated with using USVs for acoustic data collection that it does not address. Although the lack of trawl sampling data in single year is unlikely to impact our estimate given that a multi-year average of size and species composition is used in the EBS krill index (Ressler, 2022), long-term changes in krill size at the time of sampling would be missed in the continued absence of direct sampling. This is especially relevant as continued warming may lead to smaller krill, a trend that has been experimentally observed in *E. superba* (Michael et al., 2021). Similarly, as the ranges of marine species continue to shift due to climate change (Mueter and Litzow, 2008; Pinsky et al., 2013; Rooper et al., 2021), continued trawl validation of backscatter is prudent as the

composition of scattering species is unlikely to remain consistent. Autonomous platforms are also likely to influence fish and plankton behavior differently than research vessels, and species- and platform-specific reactions are likely to complicate interpretation in comparisons between autonomous platforms and the ship-based platforms that they are supplementing or replacing (Brierley et al., 2003; De Robertis et al., 2019; Guihen et al., 2022; Totland and Johnsen, 2022).

The potential for uncrewed platforms to collect acoustic observations at lower cost compared to crewed vessel-based surveys (Benoit-Bird et al., 2018; Handegard et al., 2024) provides an opportunity to extend the temporal footprint of survey efforts. For example, USVs have been used for repeat surveys examining intraseasonal trends in gadid distribution in the Chukchi Sea (Levine et al., 2020), and upward-looking moorings have been used to characterize the seasonal movements of pollock (De Robertis et al., 2018) and forage fishes (Kaltenberg et al., 2010). Uncrewed platforms can also extend the spatial footprint of surveys. USVs can also extend observations into shallower water than research vessels (Brehmer et al., 2006; Totland and Johnsen, 2022), and underwater vehicles offer an opportunity to collect high-frequency measurements (which are typically limited to a range of 100 s of meters) at depth, allowing for high-frequency acoustic measurements of deep-water organisms (Benoit-Bird et al., 2016). Additionally, many USVs (including the Saildrone USVs used in the current study; De Robertis et al., 2019; Evans et al., 2023) are smaller and quieter than standard research vessels, which can allow for acoustic observations to be made without disturbing fish that school near the surface (Komiya et al., 2024). Finally, the adoption of autonomous platforms may be the only way to continue long-term research efforts when ship-based efforts have been deemed too costly to continue (Reiss et al., 2021). In all of the above cases, is it important to ensure that the capabilities of the USV platform (for example, platform endurance/durability and operational constraints due to wind or sea state) and the echosounder (i.e. the ability to collect data at the necessary frequencies) are appropriate to the specific survey region and species (Handegard et al., 2024).

Resource managers and scientists will continue to require reliable information about species distribution and abundance over time. Unfortunately, given the challenges of funding and staffing large survey efforts, as well as the potential for disruption by unexpected events, it will remain difficult to provide this information reliably (ICES, 2020; Bryan and Thorson, 2023). Autonomous platforms have the potential to provide a practical and cost-effective way to continue or expand survey time series (Handegard et al., 2024), and will likely see increased application to supplement or replace traditional survey vessels. However, these platforms are likely to have fewer capabilities than large crewed ships. The current study provides an example of a pragmatic approach that allowed us to integrate observations from a data-limited platform into an existing time series in the face of unexpected disruptions to a ship-based survey. Responding to unexpected survey disruptions and integrating observations from new platforms in a manner that allows for existing time series to be continued will require creative solutions.

CRediT authorship contribution statement

Alex De Robertis: Writing – review & editing, Validation, Methodology, Formal analysis, Conceptualization. **Mike Levine:** Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fishres.2025.107270.

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